

Coarsening and Large-Alphabet Projections for Log Audits: Safe Partitioning Rules that Keep Certificates Sample-Feasible

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Abstract

Synthesis 25–26 provide conservative, menu-level statistical certificates for *discrete* interface projections audited from logs. In practice, many attractive projections are high-cardinality: contact-set hashes, retry traces, prefix-length profiles, or timing without bucketing. Large alphabets make finite-sample certificates vacuous.

This note centralizes one narrow tool: *coarsening* (deterministic partitioning/post-processing) as an explicit, receipt-facing design step. We record the two properties this series relies on: (i) coarsening cannot *increase* distinguishability (TV and max-divergence contract under post-processing), and (ii) coarsening reduces sample complexity by shrinking the audited alphabet. The goal is non-redundancy: mechanism and contract papers can cite this note for “how to make the audit feasible” without repeating statistics.

1 Setting

A deployment chooses an interface projection $X = \pi(T)$ of a transcript T and wants to certify (from logs) that the law of X is nearly key-independent (Anonymity B).[2] Synthesis 25 certifies discrete projections via multinomial concentration; Synthesis 27 explains how to discretize continuous timing via buckets.[3, 5] This note treats the remaining gap: what to do when the chosen discrete projection still has an alphabet that is too large to audit. For a fully wired receipt/evidence instantiation of a coarsening map (contact-set hash coarsened to a small signature), see the optional variant in Synthesis 17.[6]

2 Coarsening as a declared map

Definition 1 (Coarsening map). *Let X take values in a finite set $\mathcal{X}$. A coarsening is a deterministic map*

$$c : \mathcal{X} \rightarrow \mathcal{Y}$$

with $|\mathcal{Y}| < |\mathcal{X}|$. We write $Y := c(X)$. Receipt-grade claims are stated for Y (ENF forbids retaining/exposing X beyond what is needed to compute Y).[1]

Lemma 1 (Post-processing contracts TV and max-divergence). *For any distributions P, Q on $\mathcal{X}$ and any map c , let P^c, Q^c be the induced laws of $Y = c(X)$. Then*

$$\text{TV}(P^c, Q^c) \leq \text{TV}(P, Q), \quad D_\infty(P^c \| Q^c) \leq D_\infty(P \| Q).$$

Proof. Both are standard data-processing facts. For TV: for any event $B \subseteq \mathcal{Y}$, the preimage $c^{-1}(B) \subseteq \mathcal{X}$ is an event, hence $|P^c(B) - Q^c(B)| = |P(c^{-1}(B)) - Q(c^{-1}(B))| \leq \sup_{A \subseteq \mathcal{X}} |P(A) - Q(A)|$. For D_∞ : for any B with $Q^c(B) > 0$, $P^c(B)/Q^c(B) = P(c^{-1}(B))/Q(c^{-1}(B)) \leq \sup_{A: Q(A) > 0} P(A)/Q(A)$. $\square$

Remark 1 (What coarsening does and does not guarantee). *Coarsening is safe for privacy endpoints: if a deployment can only certify bounds for $Y = c(X)$, those bounds still upper bound distinguishability on any downstream computation of Y . However, coarsening can hide violations that would be visible at finer resolution. In this archive, that tradeoff is handled explicitly: the ENF declares what the adversary tier can see, and the contract is only about that declared surface.*[1]

3 Sample-feasibility: why $|\mathcal{Y}|$ matters

Synthesis 25 uses a Weissman-type tail bound of the form

$$\Pr [\text{TV}(\hat{P}, P) > \tau] \leq (2^m - 2) e^{-2n\tau^2}, \quad m := |\mathcal{Y}|,$$

which implies the rough budget inequality in Synthesis 27.[3, 5] Rearranging without approximation yields an explicit *alphabet constraint*:

Lemma 2 (Exact alphabet constraint). *Fix $\tau \in (0, 1)$ and failure probability $\delta \in (0, 1)$. A sufficient condition for $\Pr[\text{TV}(\hat{P}, P) \leq \tau]$ $ge 1 - \delta$ under the Weissman bound is*

$$m \leq \log_2(2 + \delta e^{2n\tau^2}).$$

Interpretation. If m is too large for the available n , a certificate for X will be too weak to be meaningful. A declared coarsening c reduces m and restores feasibility, at the cost of claiming only what is visible at the coarsened surface.

4 Design patterns (receipt-friendly)

Hierarchical coarsening (coarse $\rightarrow$ fine). Publish a stable coarse map c_0 and certify $Y_0 = c_0(X)$ routinely. Optionally, publish a refined map c_1 (or a refinement family) as a *new menu item* when enough samples are available. This keeps change control explicit: refinement consumes new audit confidence via Synthesis 26.[4]

Hash then bin (large structures). When X is a structured object (contact multiset, retry trace), deployments often first hash to a fixed domain. Audit feasibility still depends on the *effective* alphabet actually used in scope; a receipt-grade coarsening is then a binning of the hash space (e.g., “top- m ” categories plus an **other** bin). If the top- m set is data-dependent, apply Synthesis 29: either select via a published witness (state-indexed family) or separate pilot selection from frozen-audit and spend confidence on updates.[8, 4, 7]

Coarsen by declared invariants (mechanism-first). If the mechanism already enforces invariants (deadlines, prefix-length equalization classes, retry ceilings), define c so that Y is exactly the invariant class. This aligns the audit surface with mechanism feasibility analyses (stop-time normal forms, prefix capacities) and avoids “creative” partitions that are hard to review.[10, 9]

5 Receipt-facing declaration checklist

A coarsened projection should be publishable without re-deriving statistics:

- Parent projection id (ENF-ID for X) and coarsening id (map digest) for c .
- The induced alphabet size $m = |\mathcal{Y}|$ (or a conservative bound) used in the log-audit certificate.
- Statement of interface discipline: observers at the declared tier see only Y (not finer X).[1]
- Statistical evidence reference: Synthesis 25 (menu-level certificate) and, if repeated/expanding, Synthesis 26 (spending).[3, 4]

References

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- [5] Anonymous. Bucket Design for Continuous Timing Interfaces: Making “Continuous” Telemetry Auditably Discrete. Synthesis 27, The-One-True-Archive (2026).
- [6] Anonymous. Worked Example: Receipt/Artifact Interlock for OHTTP + DHT Profiling (with optional variants). Synthesis 17, The-One-True-Archive (2026).
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- [9] Anonymous. Prefix Capacities and Separation Cuts for Shared-Kernel Termination-Time Hiding. Published mathematics note (2026-01-25), The-One-True-Archive.
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